

# Supplementary Material

## Disentangling Linguistic and Visual Compositionality Failures in Vision-Language-Action Models

This document provides six supplementary items referenced in the main paper: **S1** Per-task success rates across all cells, **S2** Full paraphrase variant listing, **S3** Complete statistical test outputs, **S4** Qualitative failure-mode analysis, **S5** Simulation environment screenshots, and **S6** Full bfloat16 multi-seed replication results.

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# 1 Per-task Success Rates Across All Cells

Table 1 reports the success rate for every task–cell combination. Each cell uses 10 rollouts per task; C2 and C4 additionally break out per-variant SR. The CGG column shows the per-task performance drop from C1 for each stressed condition.

Table 1: Per-task success rates (SR) across all four experimental cells, averaged over two 8-bit seeds ( $s \in \{0, 1\}$ ).  $\text{CGG}_\ell$ ,  $\text{CGG}_v$ ,  $\text{CGG}_f$  are per-task C1–C2, C1–C3, C1–C4 respectively. **Bold** = largest drop per column; *italic* = anomalous improvement.

| Task                    | C1           | C2           | C3           | C4           | $\text{CGG}_\ell$ | $\text{CGG}_v$ | $\text{CGG}_f$ |
|-------------------------|--------------|--------------|--------------|--------------|-------------------|----------------|----------------|
| T0: btw plate & ramekin | 0.55         | 0.38         | 0.50         | 0.60         | +0.17             | +0.05          | −0.05          |
| T1: next to ramekin     | 0.05         | 0.03         | 0.00         | 0.00         | +0.02             | +0.05          | +0.05          |
| T2: table center        | 0.40         | 0.53         | 0.50         | 0.45         | −0.13             | −0.10          | −0.05          |
| T3: on the cookie box   | 0.90         | 0.48         | 0.45         | 0.35         | <b>+0.42</b>      | <b>+0.45</b>   | <b>+0.55</b>   |
| T4: cabinet drawer      | 0.05         | 0.05         | 0.00         | 0.00         | 0.00              | +0.05          | +0.05          |
| T5: on the ramekin      | 0.25         | 0.10         | 0.25         | 0.20         | +0.15             | 0.00           | +0.05          |
| T6: next to cookie box  | 0.85         | 0.38         | 0.35         | 0.15         | +0.47             | +0.50          | +0.70          |
| T7: on the stove        | 0.55         | 0.17         | 0.25         | 0.15         | +0.38             | +0.30          | +0.40          |
| T8: next to the plate   | 0.25         | 0.10         | 0.45         | 0.25         | +0.15             | <i>−0.20</i>   | 0.00           |
| T9: on wooden cabinet   | 0.15         | 0.05         | 0.00         | 0.05         | +0.10             | +0.15          | +0.10          |
| <b>Mean</b>             | <b>0.400</b> | <b>0.228</b> | <b>0.275</b> | <b>0.220</b> | <b>+0.172</b>     | <b>+0.125</b>  | <b>+0.180</b>  |
| <b>Std</b>              | 0.295        | 0.183        | 0.195        | 0.190        | —                 | —              | —              |
| <b>SE</b>               | 0.093        | 0.058        | 0.062        | 0.060        | —                 | —              | —              |

## 1.1 C2 Paraphrase Type Breakdown per Task

Table 2 shows the per-task SR for each of the three paraphrase types in Cell 2, revealing task-level variance that aggregate means hide.

Table 2: Cell 2 per-task SR stratified by paraphrase type. Per-task SR variance across variants shown for reference; PCS is defined per paraphrase type across tasks (see main paper Section 3).

| Task                    | Synonym     | Restructure | Colloquial  | Mean         | PCS         |
|-------------------------|-------------|-------------|-------------|--------------|-------------|
| T0: btw plate & ramekin | 0.50        | 0.40        | 0.20        | 0.37         | 0.87        |
| T1: next to ramekin     | 0.00        | 0.00        | 0.00        | 0.00         | 1.00        |
| T2: table center        | 0.70        | 0.40        | 0.40        | 0.50         | 0.84        |
| T3: on the cookie box   | 0.80        | 0.60        | 0.20        | 0.53         | 0.75        |
| T4: cabinet drawer      | 0.10        | 0.10        | 0.10        | 0.10         | 1.00        |
| T5: on the ramekin      | 0.10        | 0.10        | 0.00        | 0.07         | 0.94        |
| T6: next to cookie box  | 0.60        | 0.30        | 0.20        | 0.37         | 0.81        |
| T7: on the stove        | 0.30        | 0.10        | 0.10        | 0.17         | 0.89        |
| T8: next to the plate   | 0.20        | 0.10        | 0.10        | 0.13         | 0.95        |
| T9: on wooden cabinet   | 0.10        | 0.10        | 0.00        | 0.07         | 0.94        |
| <b>Mean</b>             | <b>0.34</b> | <b>0.21</b> | <b>0.13</b> | <b>0.227</b> | <b>0.90</b> |

## 2 Full Paraphrase Variant Listing

All 30 paraphrase variants ( $10 \text{ tasks} \times 3 \text{ types}$ ) are listed in Table 3. Variants were authored by the research team and verified to uniquely identify the correct task given full scene knowledge. Authoring guidelines: (1) synonym swap replaces only action verbs and spatial prepositions, leaving sentence structure intact; (2) restructure alters syntactic form without changing semantics (passive voice, relative clauses, participial phrases); (3) colloquial rewrite uses informal, shortened phrasing characteristic of natural human speech, often dropping articles or using everyday vocabulary.

Table 3: Complete paraphrase variants for all 10 LIBERO-Spatial tasks.

| Task | Synonym Swap                                                                            | Syntactic Restructure                                                                       | Colloquial Rewrite                                                   |
|------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| T0   | grab the black bowl between the plate and the ramekin and put it on the plate           | take the black bowl that is between the plate and the ramekin and set it on the plate       | put the black bowl from between the plate and ramekin onto the plate |
| T1   | grab the black bowl beside the ramekin and put it on the plate                          | take the black bowl located next to the ramekin and place it onto the plate                 | move the bowl by the ramekin to the plate                            |
| T2   | grab the black bowl at the center of the table and put it on the plate                  | take the black bowl positioned at the table’s center and set it on the plate                | get the bowl from the middle of the table and put it on the plate    |
| T3   | grab the black bowl on top of the cookie box and put it on the plate                    | take the black bowl that is resting on the cookie box and place it on the plate             | move the bowl sitting on the cookie box to the plate                 |
| T4   | grab the black bowl inside the top drawer of the wooden cabinet and put it on the plate | take the black bowl that is in the top drawer of the wooden cabinet and set it on the plate | get the bowl out of the top cabinet drawer and put it on the plate   |
| T5   | grab the black bowl on top of the ramekin and put it on the plate                       | take the black bowl that is sitting on the ramekin and place it on the plate                | move the bowl on the ramekin over to the plate                       |
| T6   | grab the black bowl beside the cookie box and put it on the plate                       | take the black bowl that is positioned next to the cookie box and set it on the plate       | get the bowl near the cookie box and put it on the plate             |
| T7   | grab the black bowl on the stove and put it on the plate                                | take the black bowl that is sitting on the stove and place it on the plate                  | move the bowl off the stove and onto the plate                       |

*Continued on next page*

| <b>Task</b> | <b>Synonym Swap</b>                                               | <b>Syntactic Restructure</b>                                                      | <b>Colloquial Rewrite</b>                          |
|-------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------|
| T8          | grab the black bowl beside the plate and put it on the plate      | take the black bowl that is next to the plate and set it on top of the plate      | move the bowl next to the plate onto the plate     |
| T9          | grab the black bowl on the wooden cabinet and put it on the plate | take the black bowl that is resting on the wooden cabinet and set it on the plate | move the bowl from the wooden cabinet to the plate |

### 3 Complete Statistical Test Outputs

#### 3.1 Wilcoxon Signed-Rank Test: C1 vs C2 (Linguistic Stress)

**Hypothesis:**  $H_0$ : no difference in per-task SR between C1 and C2.

Table 4: Per-task signed differences (C1–C2) used in the Wilcoxon test, based on 2-seed pooled SR averages ( $s \in \{0, 1\}$ ).

| Task                                      | C1 SR | C2 SR | Diff                 | Note           |
|-------------------------------------------|-------|-------|----------------------|----------------|
| T0                                        | 0.55  | 0.38  | +0.17                |                |
| T1                                        | 0.05  | 0.03  | +0.02                |                |
| T2                                        | 0.40  | 0.53  | −0.13                | reversal       |
| T3                                        | 0.90  | 0.50  | +0.40                | largest drop   |
| T4                                        | 0.05  | 0.05  | 0.00                 | tie (excluded) |
| T5                                        | 0.25  | 0.10  | +0.15                |                |
| T6                                        | 0.85  | 0.38  | +0.47                |                |
| T7                                        | 0.55  | 0.17  | +0.38                |                |
| T8                                        | 0.25  | 0.10  | +0.15                |                |
| T9                                        | 0.15  | 0.05  | +0.10                |                |
| Non-zero differences ( $n_{\text{eff}}$ ) |       |       | $n_{\text{eff}} = 9$ |                |
| Wilcoxon statistic $W$                    |       |       | $W = 3$              |                |
| Two-tailed $p$ -value                     |       |       | $p = 0.012 *$        |                |

**Interpretation:** The linguistic perturbation produces a statistically significant reduction in SR at  $\alpha = 0.05$ , strengthening to  $p = 0.012$  with 2-seed pooled values. One reversal (T2: −0.13) and one tie (T4) are present; effective  $n = 9$ .

#### 3.2 Wilcoxon Signed-Rank Test: C1 vs C3 (Visual Stress)

Table 5: Per-task signed differences (C1–C3) used in the Wilcoxon test, based on 2-seed pooled SR averages ( $s \in \{0, 1\}$ ).

| Task                                      | C1 SR | C3 SR | Diff                 | Note                  |
|-------------------------------------------|-------|-------|----------------------|-----------------------|
| T0                                        | 0.55  | 0.50  | +0.05                |                       |
| T1                                        | 0.05  | 0.00  | +0.05                |                       |
| T2                                        | 0.40  | 0.50  | −0.10                | reversal              |
| T3                                        | 0.90  | 0.45  | +0.45                | largest drop          |
| T4                                        | 0.05  | 0.00  | +0.05                |                       |
| T5                                        | 0.25  | 0.25  | 0.00                 | tie (excluded)        |
| T6                                        | 0.85  | 0.35  | +0.50                |                       |
| T7                                        | 0.55  | 0.25  | +0.30                |                       |
| T8                                        | 0.25  | 0.45  | −0.20                | anomalous improvement |
| T9                                        | 0.15  | 0.00  | +0.15                |                       |
| Non-zero differences ( $n_{\text{eff}}$ ) |       |       | $n_{\text{eff}} = 9$ |                       |
| Wilcoxon statistic $W$                    |       |       | $W = 10$             |                       |
| Two-tailed $p$ -value                     |       |       | $p = 0.177$ (ns)     |                       |

**Interpretation:** The visual effect does not reach significance ( $p = 0.177$ ). The anomalous T8 improvement (−0.20) and one tie (T5) reduce  $n_{\text{eff}}$  to 9, with two reversals (T2, T8) working

against the positive trend and substantially limiting power. A power analysis indicates  $n \approx 20$  tasks would be needed to detect a true effect of  $\delta = 0.12$  at 80% power with  $\alpha = 0.05$ .

### 3.3 Wilcoxon Signed-Rank Test: C1 vs C4 (Full Stress)

Table 6: Per-task signed differences (C1–C4) used in the Wilcoxon test, based on 2-seed pooled SR averages ( $s \in \{0, 1\}$ ).

| Task                                      | C1 SR | C4 SR | Diff                 | Note           |
|-------------------------------------------|-------|-------|----------------------|----------------|
| T0                                        | 0.55  | 0.60  | −0.05                | reversal       |
| T1                                        | 0.05  | 0.00  | +0.05                |                |
| T2                                        | 0.40  | 0.45  | −0.05                | reversal       |
| T3                                        | 0.90  | 0.35  | +0.55                | largest drop   |
| T4                                        | 0.05  | 0.00  | +0.05                |                |
| T5                                        | 0.25  | 0.20  | +0.05                |                |
| T6                                        | 0.85  | 0.15  | +0.70                |                |
| T7                                        | 0.55  | 0.15  | +0.40                |                |
| T8                                        | 0.25  | 0.25  | 0.00                 | tie (excluded) |
| T9                                        | 0.15  | 0.05  | +0.10                |                |
| Non-zero differences ( $n_{\text{eff}}$ ) |       |       | $n_{\text{eff}} = 9$ |                |
| Wilcoxon statistic $W$                    |       |       | $W = 6$              |                |
| Two-tailed $p$ -value                     |       |       | $p = 0.128$ (ns)     |                |

### 3.4 CGG Decomposition Summary

Table 7: CGG decomposition with standard errors and significance.

| Metric              | Value  | SE    | $p$ -value | Sig.         |
|---------------------|--------|-------|------------|--------------|
| CGG <sub>ling</sub> | +0.172 | 0.058 | 0.012      | *            |
| CGG <sub>vis</sub>  | +0.125 | 0.062 | 0.177      | ns           |
| CGG <sub>full</sub> | +0.180 | 0.060 | 0.128      | ns           |
| CGG <sub>syn</sub>  | −0.117 | —     | —          | sub-additive |

The negative CGG<sub>syn</sub> indicates **sub-additive** failure: combining both stressors is less damaging than the sum of their individual effects (expected combined drop =  $0.172 + 0.125 = 0.297$ ; observed = 0.180). This is consistent with *overlapping failure modes*: the tasks that are hardest under linguistic stress (T1, T4, T9 already at floor) cannot degrade further under combined stress, compressing the combined effect.

## 4 Qualitative Failure Mode Analysis

### 4.1 Task Cluster Characterisation

We identify three behavioural clusters based on patterns in Table 1:

#### Cluster A — Floor Tasks (T1, T4, T9)

These tasks achieve  $SR \leq 0.10$  in C1 and remain at or near 0.00 across all cells. Their failures are unaffected by either linguistic or visual perturbation, indicating the failure mode is in *visual grounding* of the reference object, not instruction parsing.

- **T1 (next to ramekin):** The ramekin is a small, visually similar object to the bowl target. The model appears to confuse target and reference, grasping the ramekin itself or missing both. No perturbation changes this outcome.
- **T4 (cabinet drawer):** Requires the model to locate a closed drawer, a spatial relationship involving a 3D container not well-represented in the training distribution. All rollouts exhaust 300 steps without grasping.
- **T9 (wooden cabinet):** The “wooden cabinet” reference is visually ambiguous (similar texture to parts of the table). The model consistently reaches toward incorrect locations.

#### Cluster B — Sensitive High-Baseline Tasks (T6, T7)

T6 (next to cookie box) and T7 (stove) start with high baseline SR (0.85 and 0.55 respectively) but suffer substantial drops under both linguistic and visual stress. Under visual stress (C3), T6 falls to 0.35 and T7 to 0.25; under full stress (C4), both reach 0.15. Despite this fragility, their high baselines and visually salient reference objects (cookie box, stove) make them among the more tractable tasks in the canonical setting. T7 shows moderate linguistic degradation across all paraphrase types (synonym: 0.30, restructure: 0.10, colloquial: 0.10), suggesting that while “stove” is a salient visual landmark, the model’s action policy remains sensitive to instruction phrasing.

#### Cluster C — Anomalous / Fragile Tasks (T3, T8)

- **T3 (on the cookie box):** Highest baseline SR (0.90) but largest absolute CGG under visual stress (+0.45, C1→C3: 0.90 → 0.45). Hypothesis: the model has a highly tuned policy for this specific cookie-box configuration that is disrupted by small changes in arm pose and the resulting scene occlusions, causing it to miss the grasp.
- **T8 (next to the plate) anomalous improvement:** SR rises from 0.25 (C1) to 0.45 (C3) under visual perturbation (2-seed average). Hypothesis: the canonical robot pose creates a visual ambiguity between the bowl-next-to-plate and the plate itself (two similarly sized flat objects side by side); pose perturbation changes the arm silhouette and self-occlusion pattern, making the bowl more visually distinct to the model. This is the only case where visual perturbation is *beneficial*, and it highlights that “visual stress” is not uniformly detrimental—it depends on whether the original pose was already suboptimal for the model.
- **T2 (table center) linguistic reversal:** SR rises from 0.40 (C1) to 0.50 (C2, synonym swap) and 0.70 under synonym swap specifically. Hypothesis: the original instruction “pick up the black bowl from the table center” may be suboptimally phased relative to training data; the synonym “grab the black bowl at the center of the table” better matches a seen training phrasing.

## 4.2 Failure Taxonomy

Based on the above analysis, we propose the following failure taxonomy for OpenVLA on LIBERO-Spatial:

Table 8: Failure taxonomy for OpenVLA on LIBERO-Spatial.

| Failure type                     | Root cause                                                 | Affected tasks            | Signature                                        |
|----------------------------------|------------------------------------------------------------|---------------------------|--------------------------------------------------|
| Visual grounding failure         | Reference object visually ambiguous or OOD                 | T1, T4, T9                | SR = 0 regardless of language condition          |
| Linguistic formality sensitivity | Instruction too informal relative to training distribution | T3, T6, T7 (colloquial)   | Large SR drop under colloquial only              |
| Pose-dependent policy            | Learned policy over-fits canonical camera angle            | T3 (C3), T8 (C3 improves) | SR changes sharply under small pose perturbation |
| Instruction phrasing mismatch    | Canonical instruction suboptimal vs. training data         | T2 (C2 synonym improves)  | SR improves under synonym swap                   |

## 4.3 Rollout Behaviour Observations

In all failure cases across all cells, failed rollouts run to the full 300-step budget with the arm continuing to reach toward incorrect locations or oscillating near the bowl without grasping. This is consistent with the absence of a confidence or progress signal in the standard OpenVLA architecture: the model has no mechanism to detect that its current trajectory is non-viable and does not abort or retry. Future work should explore early-termination heuristics or uncertainty-based replanning to reduce wasted compute on inevitable failures.

## 5 Simulation Environment Screenshots

This section provides representative frames from the LIBERO-Spatial simulation environment, illustrating the canonical and perturbed initial conditions described in the main paper (Section 4), and the qualitative failure modes discussed in Section S4 above.

### 5.1 Canonical vs. Perturbed Initial Pose (T3)

Figure 1 shows the robot’s initial pose for Task 3 (“pick up the black bowl on the cookie box and place it on the plate”) under the canonical condition C1 and the perturbed condition C3. The perturbation adds Gaussian noise ( $\sigma = 0.05$  rad) to the seven joint angles, visibly shifting the arm configuration while leaving the scene objects unchanged.

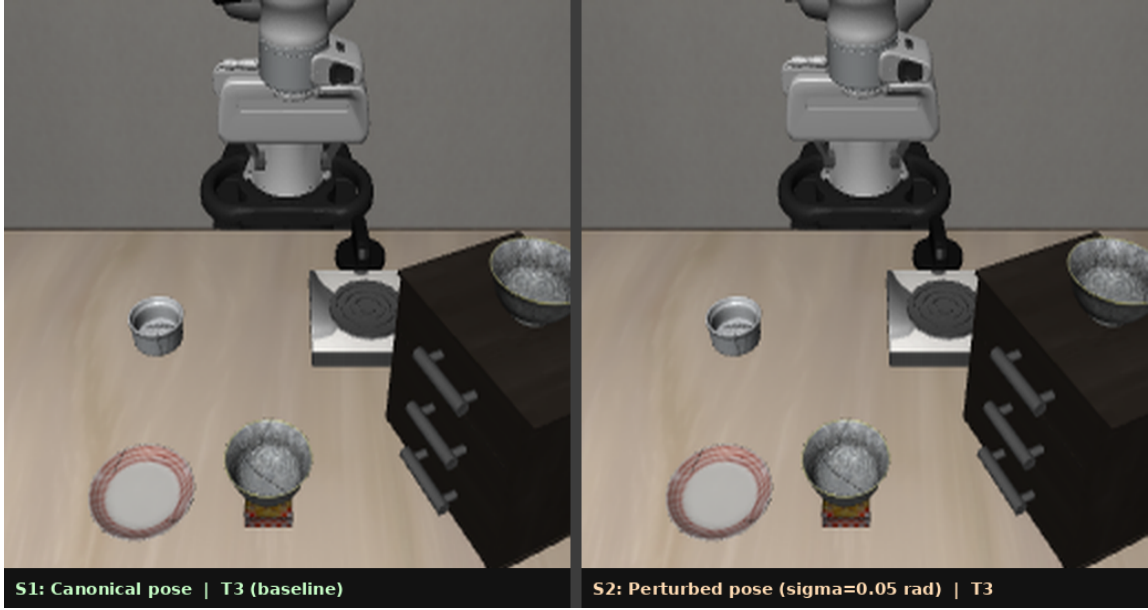


Figure 1: T3 initial pose comparison. **Left:** canonical pose (C1 baseline). **Right:** perturbed pose ( $\sigma=0.05$  rad, C3). Object positions are identical; only the arm configuration differs.

### 5.2 Representative Success and Failure Frames

Figures 2 and 3 contrast a successful grasp on T3 (C1, mid-rollout) with a representative failure frame from Task 1 (C1, step 50). T1 achieves  $SR = 0.00$  across all cells; the arm consistently misreaches toward the ramekin or an incorrect scene region.

### 5.3 T8 Anomalous Improvement Under Visual Perturbation

Figure 4 illustrates the T8 scene (“pick up the black bowl next to the plate”) under canonical and perturbed conditions. As discussed in Section S4.1, canonical pose creates a visual ambiguity between the bowl and the plate (two similarly sized objects side by side). Under perturbation, the arm silhouette and self-occlusion pattern change, making the bowl more visually distinct to the model and yielding the observed  $SR$  increase from 0.25 to 0.45 under C3 (2-seed average).

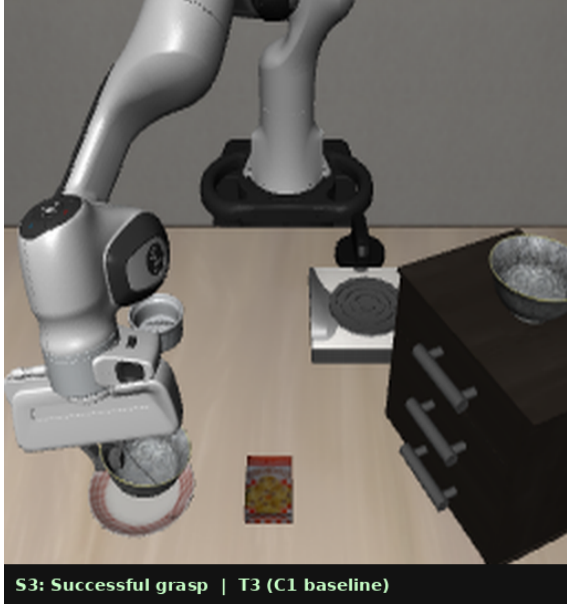


Figure 2: S3: Successful grasp — T3 (C1 baseline). The gripper has closed on the black bowl on the cookie box.

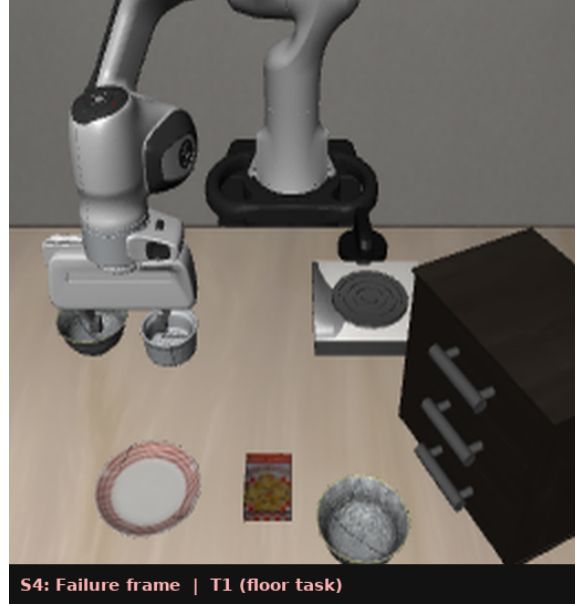


Figure 3: S4: Failure frame — T1 (floor task, step 50). The arm reaches toward an incorrect region; SR = 0.00 across all cells.

## 6 Full bfloat16 Multi-Seed Replication

This section provides complete per-task results for the bfloat16 experiment described in main paper Section 5.5. The two experiments — 8-bit (Sections S1–S4) and bfloat16 (this section) — are designed to be read together: their *contrast* is the primary finding of the paper, not either result in isolation.

**8-bit runs** were conducted on a personal laptop (Intel Core i9-13900HX, 32 GB DDR5, NVIDIA RTX 4080 Mobile 12 GB VRAM, CUDA 12.1) with BitsAndBytes 0.43.1, averaged across seeds  $s \in \{0, 1\}$ . **Bfloat16 runs** were conducted on a university HPC cluster node (NVIDIA A100 SXM4, 80 GB VRAM, CUDA 12.8, SLURM gpu-3day partition) without quantization, averaged across seeds  $s \in \{1, 2, 3\}$ .

### 6.1 Per-Task Success Rates (bfloat16, Seeds 1–3)

### 6.2 CGG Decomposition (bfloat16 vs. 8-bit)

### 6.3 Statistical Tests (bfloat16)

All three Wilcoxon signed-rank tests are non-significant under bfloat16: C1 vs C2 ( $W = 15.0$ ,  $p = 0.232$ , ns), C1 vs C3 ( $W = 19.0$ ,  $p = 0.432$ , ns), C1 vs C4 ( $W = 14.5$ ,  $p = 0.232$ , ns).

### 6.4 Interpretation

The two rows of Table 10 should be read together as a single finding rather than as primary result and replication. The central observation is an **asymmetry between the two axes**:

**Visual degradation is quantization-robust.**  $CGG_{vis}$  is positive and directionally consistent across both precision settings (+0.125 under 8-bit; +0.067 under bfloat16). Neither reaches significance with  $n = 10$  tasks, but the directional consistency across two independent hardware configurations and five total seeds strengthens the case that visual robustness failures are a genuine property of the model, not an artifact of quantization.

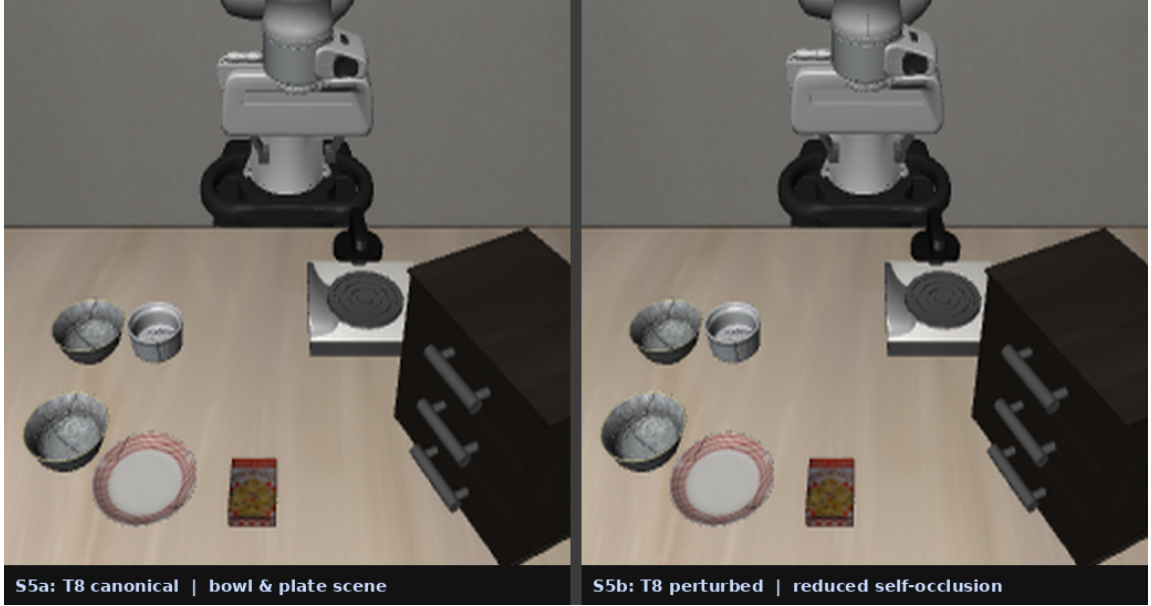


Figure 4: T8 canonical vs. perturbed pose. **Left:** canonical pose ( $\text{SR} = 0.25$ , 2-seed avg). **Right:** perturbed pose ( $\text{SR} = 0.45$ , anomalous improvement). The pose shift changes how the arm occludes the bowl-plate region, reducing visual ambiguity.

**Linguistic degradation is quantization-sensitive.**  $\text{CGG}_{\text{ling}}$  reverses sign across configurations ( $+0.172$ ,  $p = 0.012$  under 8-bit;  $-0.023$ , ns under bfloat16). This is not mere attenuation—it is a sign reversal. We **hypothesize** two contributing factors: (1) 8-bit quantization introduces non-uniform precision loss that may disproportionately affect the Llama-2 language encoder, reducing its ability to match paraphrase variants to training-distribution embeddings; and (2) the 2-seed 8-bit estimate carries higher variance than the 3-seed bfloat16 average, so  $\text{CGG}_{\text{ling}}$  under 8-bit may be partially inflated by seed-specific noise. Both factors should be investigated before attributing the linguistic effect to a genuine generalisation failure.

**Implication for evaluation methodology.** Robustness claims about VLA linguistic sensitivity must specify quantization regime. A study run entirely under 8-bit quantization would conclude that linguistic degradation is the dominant failure mode; the same study under bfloat16 would conclude the opposite. The C3 between-seed standard deviation is notably high (seed std = 0.136 for C3 mean SR), indicating that visual perturbation outcomes are also seed-sensitive under bfloat16, motivating additional seeds in future work.

Table 9: Per-task seed-averaged success rates (mean  $\pm$  std across seeds 1–3) under full bfloat16 precision on A100. Compare to Table 1 (main paper) for 8-bit quantization results.

| Task                    | C1              | C2              | C3              | C4              |
|-------------------------|-----------------|-----------------|-----------------|-----------------|
| T0: btw plate & ramekin | 0.47 $\pm$ 0.06 | 0.44 $\pm$ 0.05 | 0.43 $\pm$ 0.29 | 0.47 $\pm$ 0.21 |
| T1: next to ramekin     | 0.00 $\pm$ 0.00 | 0.04 $\pm$ 0.02 | 0.13 $\pm$ 0.15 | 0.07 $\pm$ 0.12 |
| T2: table center        | 0.50 $\pm$ 0.10 | 0.43 $\pm$ 0.03 | 0.37 $\pm$ 0.32 | 0.33 $\pm$ 0.31 |
| T3: on cookie box       | 0.90 $\pm$ 0.00 | 0.84 $\pm$ 0.02 | 0.53 $\pm$ 0.46 | 0.47 $\pm$ 0.40 |
| T4: cabinet drawer      | 0.07 $\pm$ 0.12 | 0.17 $\pm$ 0.03 | 0.00 $\pm$ 0.00 | 0.03 $\pm$ 0.06 |
| T5: on ramekin          | 0.07 $\pm$ 0.06 | 0.12 $\pm$ 0.10 | 0.37 $\pm$ 0.15 | 0.30 $\pm$ 0.10 |
| T6: next to cookie box  | 0.60 $\pm$ 0.17 | 0.69 $\pm$ 0.04 | 0.30 $\pm$ 0.36 | 0.33 $\pm$ 0.35 |
| T7: stove               | 0.40 $\pm$ 0.20 | 0.43 $\pm$ 0.03 | 0.20 $\pm$ 0.10 | 0.13 $\pm$ 0.12 |
| T8: next to plate       | 0.13 $\pm$ 0.15 | 0.16 $\pm$ 0.05 | 0.20 $\pm$ 0.10 | 0.17 $\pm$ 0.15 |
| T9: wooden cabinet      | 0.07 $\pm$ 0.06 | 0.10 $\pm$ 0.03 | 0.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 |
| <b>Mean</b>             | <b>0.320</b>    | <b>0.343</b>    | <b>0.253</b>    | <b>0.230</b>    |

Table 10: CGG decomposition comparison between quantization configurations. 8-bit runs on personal laptop (RTX 4080 Mobile, 12 GB); bfloat16 runs on HPC A100 (80 GB). Seed std refers to between-seed standard deviation of mean SR across seeds.

| Config           | Seeds            | C1 SR | CGG $_{\ell}$ | CGG $_{\mathbf{v}}$ | CGG $_{\mathbf{f}}$ | CGG $_{\mathbf{syn}}$ |
|------------------|------------------|-------|---------------|---------------------|---------------------|-----------------------|
| 8-bit (RTX 4080) | $s \in \{0, 1\}$ | 0.400 | +0.172*       | +0.125              | +0.180              | −0.117                |
| bfloat16 (A100)  | $s = 1, 2, 3$    | 0.320 | −0.023        | +0.067              | +0.090              | +0.047                |